Flame Tests: Identifying Metal Ions by Color

U2 / S2.5

NAME _____ DATE _____ PERIOD _____

GROUP MEMBERS _____

GOALS

- Run a flame test safely: one soaked stick per test, tip in the flame for a few seconds.
- Record the flame color of ten known compounds, and compare each one with its expected color.
- Identify three unknowns by matching their flame colors to the known compounds.
- Explain each color with ground and excited states: electrons absorb energy, drop back, and give off light.

PURPOSE

You will hold thirteen solutions in a flame and record the color each one gives. Ten are known compounds. The other three are unknowns, and the colors you record are the only evidence you will have to name them.

BACKGROUND

Heat excites electrons. In the flame, electrons in the metal ions absorb energy and jump from their ground state to higher energy levels. That excited state does not last. Each electron drops back down and gives off the energy as a photon of light.

The size of the drop sets the color. A photon's energy equals the gap between the two levels. A bigger drop gives more energy and a shorter wavelength, toward violet. A smaller drop gives a longer wavelength, toward red. Each metal has its own set of energy levels, so each gives off its own mix of colors.

The color comes from the metal ion. Sodium nitrate and sodium chloride both give a yellow flame, because the sodium ion is doing the work. That is why the table below lists ions, not whole compounds. Two cautions: the wooden stick itself burns yellow-orange once it catches, so watch the **first two or three seconds**; and a color is a clue, not proof, because two ions can give nearly the same color.

Not all light is visible. Your eyes only detect a narrow band of wavelengths, from violet (about 400 nm) to red (about 700 nm). A drop with a bigger energy gap than violet gives off ultraviolet light, shorter than 400 nm, and a smaller gap than red gives off infrared, longer than 700 nm. Both are real photons, and both are invisible. Magnesium is the example today: its electrons are excited and drop back like every other metal's, but most of the light it gives off is ultraviolet, so a magnesium compound adds little or no color you can see.

ION IN THE COMPOUND	SYMBOL	TYPICAL FLAME COLOR
Lithium	Li^+	Crimson red
Sodium	Na^+	Yellow
Potassium	K^+	Lilac
Calcium	Ca^{2+}	Orange-red
Strontium	Sr^{2+}	Red
Barium	Ba^{2+}	Pale green
Copper(II)	Cu^{2+}	Blue-green
Magnesium	Mg^{2+}	Little or none: mostly ultraviolet

Iron is not on this table: predict its color, then record what you actually see.

SAFETY

- Splash goggles go on before any burner is lit and stay on until every station is cleaned up, including while other groups are still testing.
- Long hair is tied back and loose sleeves are pushed up before you reach toward a flame. The flame on a burner can be nearly invisible in a bright room. Assume it is lit.
- A stick left in the flame too long catches fire. Hold only the dry end, keep the tip in the flame for just a few seconds, and if the tip keeps burning, drop it in the sink. Never set a burning or used stick on the bench.
- The nitrates on the list (sodium, potassium, calcium, barium, copper and iron nitrate) are oxidizers: they make wood and paper burn faster. Keep paper and notebooks away from the flame, and never hold a stick over or near the beakers.
- Several solutions are harmful if swallowed or if they touch your skin or eyes, especially the barium and copper compounds; boric acid (station 3) may harm fertility or an unborn child. Hold only the dry end of a stick, and never taste, smell or touch a solution. If one gets on your skin, rinse it with running water and tell me. In your eyes: go straight to the eyewash, hold your eyelids open for at least 15 minutes, and send a group member for me immediately.
- Do not pick up or move a beaker. If one tips over, step back and tell me. Do not wipe it up yourself.
- Report every spill, including water. Water on the floor is how people fall.

PRE-LAB — DUE AT THE START OF THE PERIOD

Answer these in your lab notebook, numbered 1 through 9, before you walk in. Do not write on this handout. No finished pre-lab, no burner.

1. Use the words **ground state** and **excited state** to describe what happens to an electron in the flame, from the moment it absorbs energy to the moment light comes out.
2. Sodium nitrate and sodium chloride give the same flame color. What does that tell you about which part of a compound makes the color?
3. Look at the reference table. Name the two ions whose colors are hardest to tell apart, and explain why that is a problem when you are identifying an unknown.
4. In the flame, magnesium's electrons are excited and drop back down, yet you will see almost no color. Use the Background to explain how both can be true, and name the kind of light magnesium mostly gives off.
5. Read Part B, steps 1–2. Where does your stick come from, which end do you hold, and where does the tip go?
6. Read Part B, step 3. Why do you record the color from the first two or three seconds and not after that?
7. Read Part B, steps 4 and 5. What happens to a stick after one test, and why does a stick never go back into a beaker?
8. Your stick keeps burning. Say exactly what you do, and write where the eyewash is in this room.
9. Copy Table 1 into your notebook in ink: one row per sample, in station order. Fill in **Expected color** from the reference table before you walk in (boric acid: green; iron and the unknowns: blank).

TABLE 1 · FLAME TEST RESULTS (13 ROWS: ONE PER SAMPLE, IN STATION ORDER)

Compound (name and formula)	Expected color	Actual color
Lithium chloride, LiCl		
Sodium nitrate, $NaNO_3$		
… one row for each sample, in station order …		
Unknown C		

MATERIALS

At each of the six stations: Bunsen burner and striker · two labeled beakers of solution, each with wooden stir sticks already soaking in it (one stick per student) · nothing else on the bench but your lab notebook, closed and away from the flame.

STATION	SAMPLE 1	SAMPLE 2
1	Lithium chloride, $LiCl$	Sodium nitrate, $NaNO_3$
2	Potassium nitrate, KNO_3	Calcium nitrate, $Ca(NO_3)_2$
3	Strontium chloride, $SrCl_2$	Barium nitrate, $Ba(NO_3)_2$ and a third beaker: Boric acid, H_3BO_3
4	Copper(II) nitrate, $Cu(NO_3)_2$	Magnesium chloride, $MgCl_2$
5	Iron(III) nitrate, $Fe(NO_3)_3$	Unknown A
6	Unknown B	Unknown C

PROCEDURE

Part A — Set up at your first station

1. Go to the station I assign. Goggles on, hair tied back, sleeves up.
2. Light the burner and adjust the air until the flame is steady and blue with a visible inner cone. A yellow, smoky flame needs more air.
3. Read both labels. Find those two rows in your table before you test anything.

Part B — Test each solution

ABOUT 5 MINUTES PER STATION

1. Each student takes one soaked stick from the first beaker. Lift it straight out and hold it by the dry end. Do not stir or tap it against the beaker.
2. Hold the wet tip at the edge of the flame, just above the blue inner cone.
3. Watch the color the instant the tip enters the flame. Record it in the **Actual color** column before you do anything else.
   - After two or three seconds the wood itself starts to burn yellow-orange. That color is the stick, not the solution. Do not record it.
   - Each person tests and records their own sample. Compare colors with your group, but write what you saw.
4. Drop the used stick in the sink. Never put a stick back into either beaker, even if it did not light: it carries soot and the other metal into the solution, and every group after you sees the wrong color.
5. Repeat steps 1–4 with a stick from the second beaker. At station 3, repeat again with the third beaker, boric acid, and compare its green with barium's.
6. Leave the burner lit. When I call "rotate," move to the next station number (station 6 moves to station 1). Do not move early.

Part C — The unknowns

1. Unknowns A, B and C are tested at stations 5 and 6 exactly like the others. Each one is a compound you have already tested.
2. Record the actual color in the unknown's row. Leave the identification for the post-lab.

DATA TABLES

You built Table 1 in your pre-lab. Have it open and ready before the first stick goes into the flame, and fill in each color **as you see it**, not at the end of the period.

Describe the color you see, not the color you expect. "Red" is not enough when there are two reds on the table: say crimson, bright red or orange-red, and whether it was strong or faint. If you write something wrong, draw one line through it and write the correction beside it.

DISPOSAL AND CLEANUP — LAST FIVE MINUTES

1. Every used stick goes in the sink, never the trash can, where a smoldering stick could start a fire. I collect them from the sink after class.
2. The beakers stay at their stations, exactly where they were. Do not pour any solution out, down the drain or anywhere else. I collect and store them.
3. Turn the gas off at the valve, then leave the burner where it is until it is cool to touch. Leave any sticks still soaking where they are.
4. **Wash all glassware with the Alconox solution in the blue spray bottle.** Spray inside, scrub with a brush, and rinse thoroughly — soap left in a beaker changes the next measurement made in it.
5. Rinse with distilled water and invert everything on the rack to dry.
6. Wipe your bench, push in your stool, and wash your hands. **Goggles come off last**, once the whole room is finished — not once you are.

POST-LAB ANALYSIS

Answer these in your lab notebook, numbered 1 through 4.

1. Identify unknowns A, B and C. For each one, name the known compound it matches and the color evidence from your table. PART C
2. Pick one unknown that could have matched two different known compounds. Name both, explain why the flame color alone could not settle it, and say what you would use to decide. REFERENCE TABLE
3. Describe what you saw for magnesium chloride. Explain why the flame showed little or no color even though its electrons were being excited and dropping back, and say where on the spectrum most of magnesium's light is. BACKGROUND
4. Find one compound whose actual color did not match its expected color, or iron, which had no expected color. Give a specific reason from the procedure that could explain what you saw. "Human error" is not a reason. PART B